

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26189**
- **Problem Statement Title - AI-Powered Criminal Network Analysis System**
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID - SIH2026-ST-842**
- **Team Name (Registered on portal) – Stryker**



IDEA / SOLUTION :

SENTINEL: AI-Powered Intelligence Platform for Real-Time Criminal Network Analysis & Crime Deterrence

- **Unified Knowledge Graph** : Fuses police FIRs, phone call records (CDRs), bank accounts, and video feeds into one real-time network.
- **Kingpin & Boss Discovery** : Graph centrality algorithms automatically uncover covert syndicate leaders hiding behind operatives.
- **Hawala & AML Detection** : Detects 4-node circular money loops (<72h) and smurfing transfers below ₹50,000 threshold.
- **Burner Phone Triangulation** : Tracks shared mobile handsets (IMEI reuse) and co-located nocturnal meetings (<500m).
- **Women Safety Edge AI** : Live CCTV monitoring detects stalking and cornering with 1-click Dial 112 PCR dispatch.
- **Statutory Evidence Vault** : Cryptographic SHA-256 blockchain ledger ensuring 100% Section 63 BSA / 65B IEA court admissibility.

PROBLEM RESOLUTION :

❖ **Breaks Data Silos :**

Police usually examine FIRs, phone logs, and bank statements in isolation. SENTINEL automatically merges fragmented clues into an interactive 360° map in seconds.

❖ **Exposes Hidden Syndicate Bosses :**

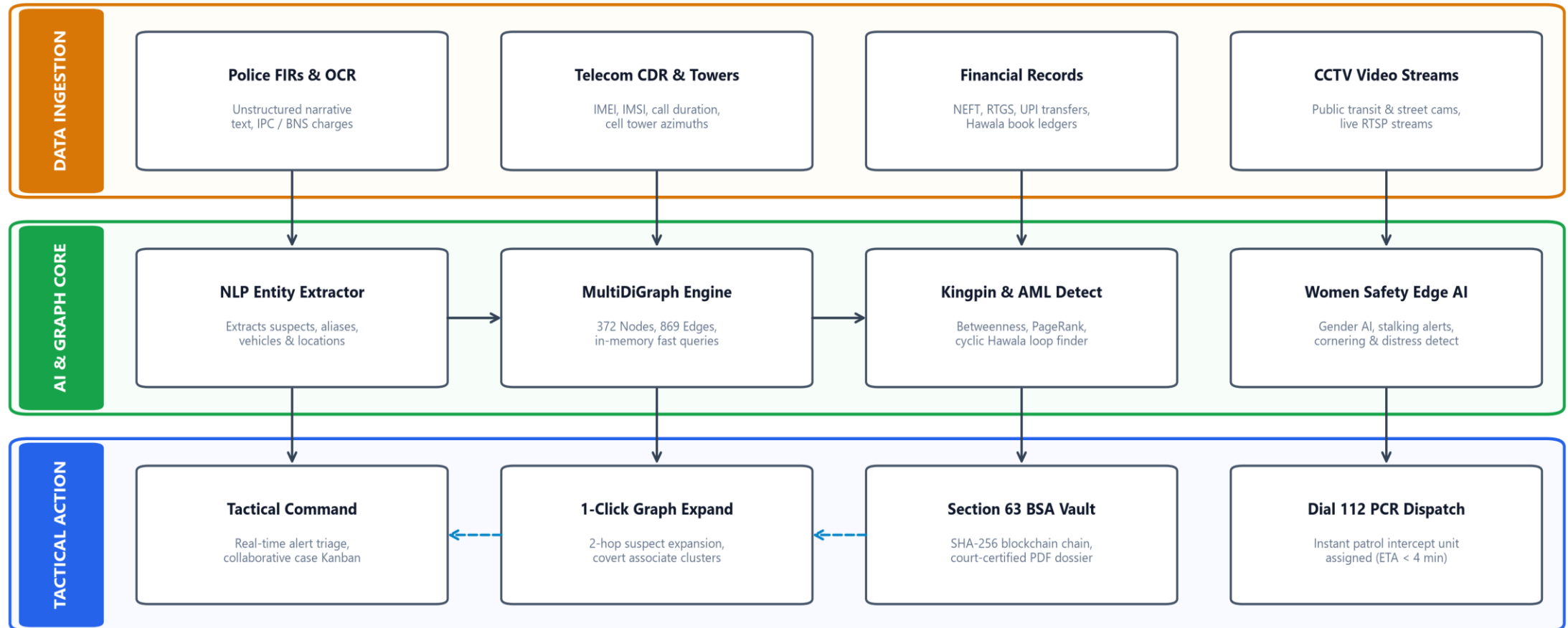
Crime kingpins rarely use their own phones. Advanced Betweenness Centrality and PageRank expose the mastermind brokers coordinating criminal activities.

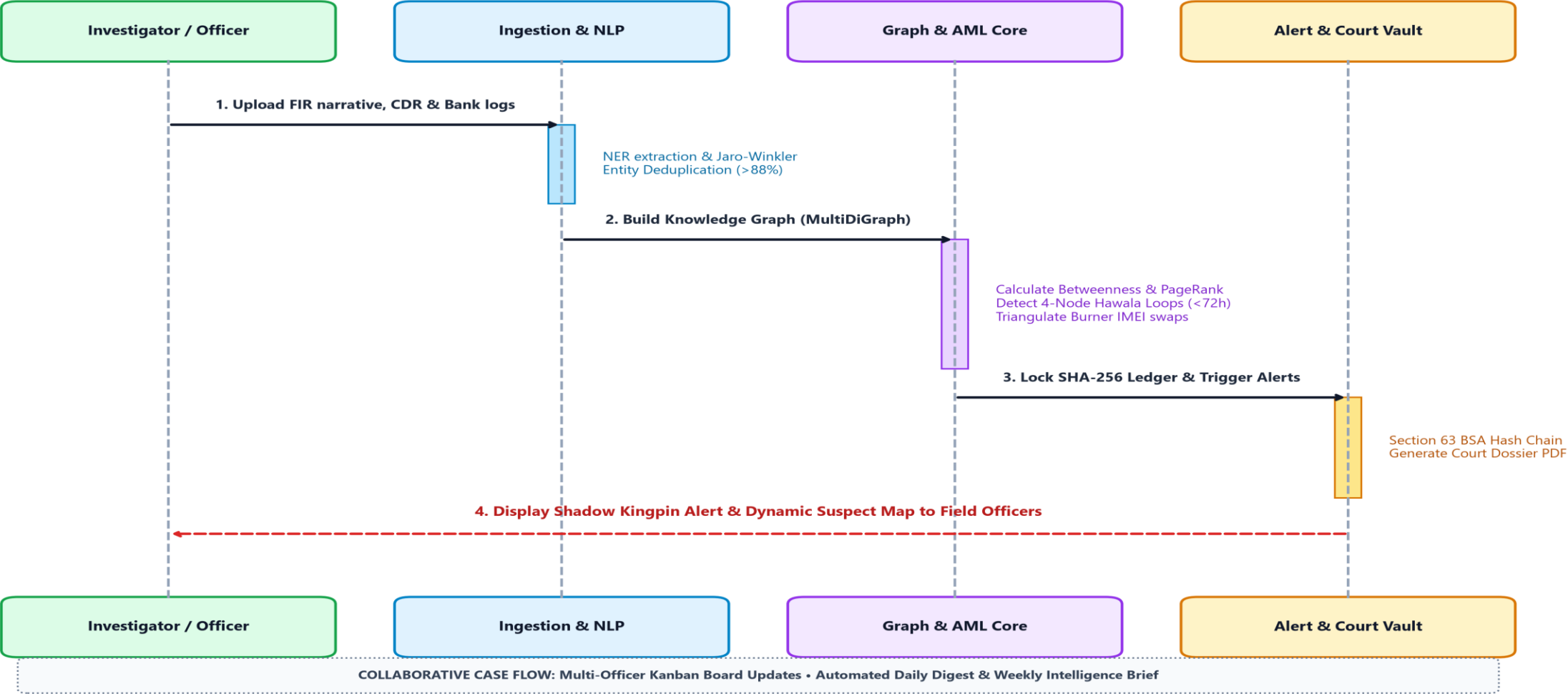
❖ **Stops Crime in Real-Time :**

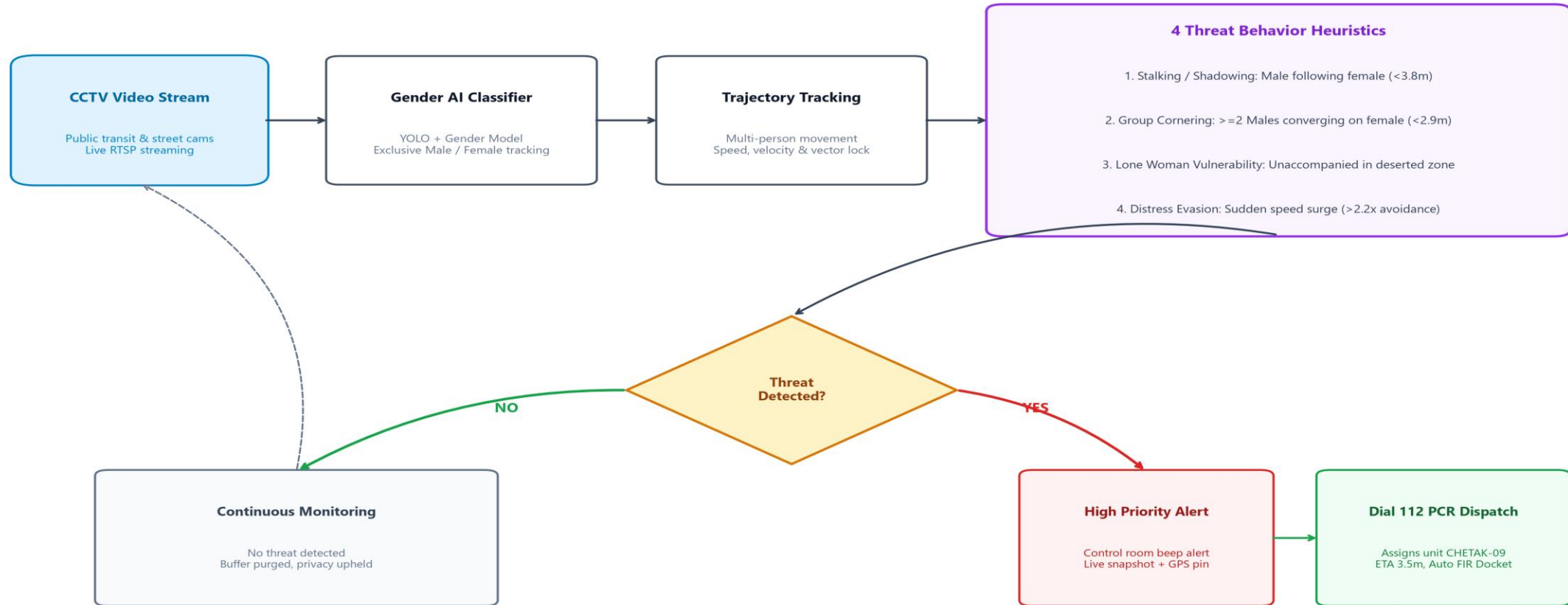
Shifts law enforcement from slow post-crime investigation to proactive live deterrence—flagging threats and dispatching emergency PCR vans in < 4 minutes.

❖ **Guarantees Court Convictions :**

Digital evidence often fails in court due to broken chain of custody. SENTINEL creates Section 63 BSA certified PDF dossiers with SHA-256 blockchain verification.







Frontend Dashboard :

- **React 19, TypeScript, Vite** – High-performance responsive tactical UI built for police control room command.
- **Tailwind CSS** – Low-light dark tactical theme reducing officer eye fatigue.
- **Cytoscape.js & Leaflet.js** – Interactive 2D/3D suspect graph networks and geo-spatial cell tower tracking.

Backend + Intelligence Core :

- **FastAPI (Python 3.12)** – Ultra-fast asynchronous REST APIs with sub-millisecond query latency.
- **NetworkX MultiDiGraph** – In-memory graph engine managing 372 nodes, 869 edges and centrality metrics.
- **SQLite WAL + Blockchain Hash Chain** – Thread-safe data storage with SHA-256 evidence sealing.
- **YOLO & spaCy NLP** – Real-time gender classification edge AI & automated FIR entity extraction.

Statutory Legal Vault :

- **ReportLab Forensic Engine** – Generates court-certified Section 63 BSA dossiers with QR code verification.

React 19

Vite & Tailwind

TypeScript

Type-safe Frontend

Python 3.12

Async Core

FastAPI

High-speed REST API

NetworkX

MultiDiGraph Analytics

Leaflet.js

Geo-Intel Tracking

SQLite WAL

Blockchain Ledger

ReportLab

Sec 63 BSA PDF

Technical Feasibility

- Uses proven open-source frameworks – FastAPI, React 19, NetworkX
- Direct compatibility with police CCTNS and telecom CDR exports
- Edge AI runs efficiently on existing workstation CPUs / GPUs
- Cryptographic SHA-256 ledger ensures 100% zero-gas data integrity

Economic Feasibility

- Zero expensive software licenses or cloud vendor lock-in
- Directly leverages city CCTV cameras & standard police PC hardware
- Compresses weeks of manual investigative analysis into seconds
- Drastically reduces police manpower costs during major syndicate probes

“The solution is technically robust, low-cost, and operationally practical—enabling police to dismantle syndicates faster while protecting vulnerable citizens.”

Challenges & Strategies

- **Unstructured Police Data** – FIRs contain spelling mistakes, slang, and varied dialects. **Solved via AI Named Entity Recognition + Jaro-Winkler fuzzy deduplication (>88%).**
- **Massive CDR & Financial Scale** – Correlating millions of call records without lag. **Solved via in-memory graph caching & SQLite WAL indexing for sub-millisecond lookups.**

Law Enforcement & Police Officers

- 1-click network expansion reveals 2-hop associates, burner phones, and money trails.
- Automatically identifies shadow kingpins who never make direct phone calls.
- Collaborative case board eliminates inter-departmental communication delays.

Women & Vulnerable Citizens

- Autonomous CCTV monitoring detects stalking, cornering, and distress in deserted transit zones.
- 1-click Dial 112 PCR dispatch sends emergency patrol units within 3.5 minutes.
- Proactive crime deterrence prevents harassment before physical harm occurs.

Judiciary & Legal Prosecution

- 100% tamper-evident digital evidence meeting Section 63 BSA / 65B IEA standards.
- Court-ready forensic PDF dossiers accelerate trials and eliminate chargesheet delays.

“SENTINEL transforms fragmented crime records into real-time operational clarity—protecting vulnerable citizens, disrupting criminal syndicates, and accelerating the delivery of justice.”

THANK YOU

SENTINEL v2.0 – AI-Powered Criminal Network Analysis System

Team Stryker • Problem Statement ID: 26189

Open for Questions & Live System Demonstration